

Clinical Management Guidelines

Emergency Medicine & Internal Medicine

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These guidelines are intended for use by medical students, junior doctors, and clinical staff at Tygerberg Academic Hospital. They summarise current evidence-based protocols for the assessment, diagnosis, and management of common acute and chronic conditions encountered in the emergency department and general medical wards. Clinical judgment must always accompany the use of any protocol.

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1. Sepsis and Septic Shock

Definition

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection (Sepsis-3 criteria, 2016). Organ dysfunction is identified by an acute change in total SOFA score of 2 points or more consequent to the infection.

Septic shock is a subset of sepsis in which circulatory and cellular/metabolic abnormalities are profound enough to substantially increase mortality. Clinically identified by the need for vasopressor therapy to maintain MAP ≥ 65 mmHg and serum lactate > 2 mmol/L in the absence of hypovolaemia.

Diagnostic Criteria

Quick SOFA (qSOFA) — screen for sepsis outside the ICU if 2 or more of:

- Respiratory rate ≥ 22 breaths/min
- Altered mentation (GCS < 15)
- Systolic blood pressure ≤ 100 mmHg

***CLINICAL NOTE:** qSOFA is a screening tool, not a diagnostic criterion. A positive qSOFA warrants full SOFA assessment and blood cultures before initiating antimicrobial therapy.*

Management Protocol — Hour-1 Bundle

Step	Intervention	Target / Notes
1	Measure serum lactate	Re-measure if initial lactate > 2 mmol/L
2	Blood cultures	Before antimicrobials; minimum 2 sets
3	Broad-spectrum antimicrobials	Within 1 hour of recognition
4	IV crystalloid resuscitation	30 mL/kg for lactate ≥ 4 mmol/L or hypotension
5	Vasopressors if hypotensive	Noradrenaline; target MAP ≥ 65 mmHg

Antimicrobial Selection

Empiric therapy should be broad-spectrum and cover the most likely pathogens based on the suspected source of infection. De-escalate once culture and sensitivity results are available, typically at 48–72 hours.

- **Unknown source:** Piperacillin-tazobactam 4.5 g IV 6-hourly + Gentamicin 5–7 mg/kg IV daily
- **Urinary source:** Ceftriaxone 2 g IV daily; adjust per culture
- **Intra-abdominal:** Metronidazole 500 mg IV 8-hourly + Ceftriaxone 2 g IV daily
- **Suspected MRSA:** Add Vancomycin; target trough 15–20 mg/L

2. Acute Hypertensive Crisis

Classification

Hypertensive crisis is defined as a systolic blood pressure (SBP) >180 mmHg and/or diastolic blood pressure (DBP) >120 mmHg. It is classified as either a hypertensive **urgency** (no acute end-organ damage) or a hypertensive **emergency** (acute end-organ damage present).

Feature	Hypertensive Urgency	Hypertensive Emergency
Blood pressure	>180/120 mmHg	>180/120 mmHg
End-organ damage	Absent	Present
Symptoms	Headache, anxiety	Chest pain, dyspnoea, confusion, focal neurology
Setting	Outpatient / ED	ICU or High Care
Treatment	Oral antihypertensives; reduce BP over 24–48 h	IV antihypertensives; reduce MAP by ≤25% in first hour

Intravenous Antihypertensive Agents

Drug	Dose	Onset	Indication
Labetalol	20–80 mg IV bolus q10 min; or 2 mg/min infusion	5 min	Most emergencies; safe in pregnancy
Hydralazine	5–10 mg IV q20 min (max 20 mg)	10 min	Eclampsia / pregnancy
Esmolol	500 mcg/kg bolus, then 50–200 mcg/kg/min	1 min	Aortic dissection, perioperative
Nicardipine	5 mg/h; titrate by 2.5 mg/h q5–15 min	1–5 min	Neurological emergencies, stroke
GTN (NTG)	5–200 mcg/min infusion	2–5 min	Acute coronary syndrome, APO

CLINICAL NOTE: Do not reduce blood pressure too rapidly in hypertensive emergency. An MAP reduction exceeding 25% in the first hour risks cerebral and coronary hypoperfusion, particularly in patients with chronic hypertension who have adapted their cerebral autoregulation.

3. Community-Acquired Pneumonia (CAP)

Severity Assessment — CURB-65 Score

Assign 1 point for each criterion present. Use to guide disposition and antimicrobial route.

Criterion	Score
C — Confusion (new disorientation)	1
U — Urea ≥ 7 mmol/L	1
R — Respiratory rate ≥ 30 breaths/min	1
B — SBP < 90 or DBP ≤ 60 mmHg	1
65 — Age ≥ 65 years	1

CURB-65 0–1: Low severity. Consider outpatient oral antibiotics. Amoxicillin 1 g 8-hourly x 5 days.

CURB-65 2: Moderate severity. Consider short inpatient stay or supervised outpatient therapy.

CURB-65 3–5: High severity. Admit. IV antibiotics. Consider HDU or ICU if score ≥ 4 .

Empiric Antimicrobial Therapy

- **Non-severe CAP (outpatient):** Amoxicillin 1 g PO 8-hourly x 5 days. If atypical cover needed, add Azithromycin 500 mg PO daily x 3 days.
- **Moderate CAP (inpatient, non-ICU):** Amoxicillin-clavulanate 1.2 g IV 8-hourly + Azithromycin 500 mg IV/PO daily.
- **Severe CAP (ICU):** Ceftriaxone 2 g IV daily + Azithromycin 500 mg IV daily. Consider Oseltamivir if influenza suspected.
- **Aspiration pneumonia:** Amoxicillin-clavulanate 1.2 g IV 8-hourly or Piperacillin-tazobactam 4.5 g IV 6-hourly.

4. Diabetic Ketoacidosis (DKA)

Diagnostic Criteria

- Blood glucose >11 mmol/L (or known diabetes mellitus)
- Blood ketones ≥ 3 mmol/L or urine ketones 2+ or more
- Venous pH <7.3 and/or bicarbonate <15 mmol/L

Fluid Resuscitation

Use 0.9% sodium chloride (normal saline) for initial resuscitation. Switch to 0.45% sodium chloride with 20 mmol/L KCl once the patient is haemodynamically stable.

Time	Fluid	Rate
0–60 min	0.9% NaCl	1 L over 1 hour (if hypotensive, 500 mL bolus first)
1–2 h	0.9% NaCl + 20 mmol KCl	500 mL/hour
2–4 h	0.9% NaCl + 20 mmol KCl	250 mL/hour
4–8 h	0.45% NaCl + 20 mmol KCl	125–250 mL/hour
≥ 8 h	0.45% NaCl + 20 mmol KCl	Titrate to clinical response

Insulin Protocol

Do NOT start insulin until potassium ≥ 3.5 mmol/L. Hypokalaemia in the context of insulin administration is a life-threatening complication.

- Fixed rate IV insulin infusion (FRIII): 0.1 units/kg/hour of Actrapid in 0.9% NaCl
- Target glucose fall: 3–6 mmol/L per hour
- Switch to variable rate insulin infusion (VRII) once ketones <0.6 mmol/L, pH >7.3 , and patient is eating and drinking
- Restart long-acting subcutaneous insulin at least 30 minutes before stopping IV insulin

CLINICAL NOTE: Potassium replacement is mandatory in DKA. Insulin drives potassium intracellularly. Check potassium every 1–2 hours in the first 4 hours. Replace aggressively via IV route if below 3.5 mmol/L.

5. Acute Coronary Syndrome (ACS)

Classification

- **STEMI** — ST-elevation myocardial infarction. Complete occlusion of coronary artery. Primary PCI within 90 minutes of first medical contact.
- **NSTEMI** — Non-ST-elevation myocardial infarction. Elevated troponin without persistent ST-elevation. Manage medically; early invasive strategy within 24–72 hours.
- **Unstable Angina** — Ischaemic symptoms at rest or minimal exertion without troponin rise. Admit for monitoring and risk stratification.

Initial Management — MONA + Antiplatelet Therapy

Drug	Dose	Notes
Morphine	2–4 mg IV PRN	For uncontrolled ischaemic pain only; may mask symptoms
Oxygen	Only if SpO ₂ <94%	Avoid hyperoxia — worsens infarct size
GTN	0.4 mg SL; or IV infusion	Avoid if hypotensive or RV infarction suspected
Aspirin	300 mg PO loading dose	Chewed, not swallowed whole; then 75–100 mg daily
Ticagrelor	180 mg PO loading dose	Preferred P2Y ₁₂ inhibitor; then 90 mg BD
Clopidogrel	300–600 mg PO loading dose	Alternative if Ticagrelor not available
Heparin	60 units/kg IV bolus (max 4000 u)	UFH infusion 12 units/kg/h (max 1000 u/h)

6. Drug Dosing Reference

The following table summarises common medications used in the emergency department and medical wards at Tygerberg. Doses are for adult patients with normal renal and hepatic function unless otherwise stated. Always verify against current formulary and local antimicrobial stewardship guidelines before prescribing.

Drug	Class	Indication	Adult Dose	Route	Cautions
Amoxicillin	Penicillin antibiotic	CAP, UTI, soft tissue	500 mg–1 g 8-hourly	PO / IV	Penicillin allergy
Ceftriaxone	3rd-gen cephalosporin	CAP, sepsis, meningitis	1–2 g daily	IV / IM	Hypersensitivity
Metronidazole	Nitroimidazole	Anaerobic infection, C. diff	500 mg 8-hourly	PO / IV	Avoid alcohol
Vancomycin	Glycopeptide	MRSA, severe Gram+	15–20 mg/kg 8–12 h	IV	Nephrotoxicity; monitor trough
Gentamicin	Aminoglycoside	Gram-neg sepsis (empiric)	5–7 mg/kg once daily	IV	Nephrotoxicity; ototoxicity
Furosemide	Loop diuretic	Acute pulmonary oedema	40–80 mg stat	IV	Electrolytes; renal function
Atenolol	Beta-blocker	Hypertension, ACS	25–100 mg daily	PO	Asthma, bradycardia, AV block
Amlodipine	Calcium channel blocker	Hypertension	5–10 mg daily	PO	Peripheral oedema
Ramipril	ACE inhibitor	Hypertension, CCF, CKD	2.5–10 mg daily	PO	Hyperkalaemia, renal artery stenosis
Warfarin	Anticoagulant	AF, DVT, PE, valve prostheses	Titrate to INR 2–3	PO	Multiple drug interactions; INR monitoring
Heparin (UFH)	Anticoagulant	ACS, PE, DVT, bridging	See ACS/PE protocol	IV	Bleeding; monitor APTT
Insulin (Actrapid)	Short-acting insulin	DKA, hyperglycaemia	See DKA protocol	IV / SC	Hypoglycaemia, hypokalaemia
Dexamethasone	Corticosteroid	Meningitis, inflammation	0.15 mg/kg 6-hourly	IV	Immunosuppression; glucose elevation
Salbutamol	Beta-2 agonist	Acute asthma, bronchospasm	2.5–5 mg nebulised PRN	Inhaled / IV	Tachycardia, hypokalaemia at high dose
Paracetamol	Analgesic / antipyretic	Pain, fever	1 g 6-hourly (max 4 g/day)	PO / IV	Hepatotoxicity in overdose; check weight
Morphine	Opioid analgesic	Severe pain, ACS	2–10 mg titrated	IV / SC / PO	Respiratory depression; constipation
Ondansetron	5-HT3 antagonist	Nausea, vomiting	4–8 mg 8-hourly	PO / IV	QTc prolongation
Midazolam	Benzodiazepine	Procedural sedation, seizure	0.05–0.1 mg/kg	IV	Respiratory depression; reversal with flumazenil